

Emotion Detection from Music Signals using CRNN and Application in Mental Health Monitoring

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Abstract—In this paper, an advanced Music Emotion Recognition (MER) system is developed using Convolutional Recurrent Neural Networks (CRNN). Utilizing the Emotify dataset [1], [2], the model was trained to classify audio clips into four emotional categories: Energy/Joy, Calm/Relax, Sadness/Nostalgia, and Tension/Fear. The methodology involves converting audio to Mel-spectrograms and applying normalization techniques (Min-Max Scaling), which proved crucial in preventing model collapse. The results demonstrated high discriminative ability, achieving an overall F1-score of 0.92, a highly competitive performance. Finally, a mental health monitoring application is proposed, which analyzes the user’s musical preferences and intervenes with supportive messages upon detecting prolonged states of melancholy.

Index Terms—Music Emotion Recognition (MER), CRNN, Deep Learning, Mel-spectrograms, Mental Health Awareness.

I. INTRODUCTION

Automatic Music Emotion Recognition (MER) is one of the most challenging fields in Music Information Retrieval (MIR). The complexity of the problem lies in the so-called “semantic gap”, which describes the disparity between the low-level acoustic features of a signal and the high-level subjective perception of emotion by humans [5].

In the international literature, early approaches relied on traditional machine learning algorithms (SVM, Random Forests). The advent of Deep Learning, particularly Convolutional Neural Networks (CNNs), allowed for more effective feature extraction. However, Choi et al. [4] highlighted that music requires models with “memory,” introducing the CRNN architecture which combines CNN layers for spatial analysis and RNN/GRU layers for temporal modeling.

In this work, a comprehensive approach based on the Emotify dataset [1] is proposed. The study focuses on the evolution of the architecture from simple CNNs to hybrid CRNN models, analyzing critical stages such as data normalization (Min-Max Scaling). The innovation of this proposal lies in utilizing the predictions to protect the user’s mental health, triggering supportive interventions in real-time.

The model’s ability to distinguish emotions is largely based on detecting the structural elements of music, such as musical scales. Historically, major scales correlate with

positive emotions (joy, energy), while minor scales are associated with darker or melancholic states (sadness, tension) [3]. The proposed CRNN model, through the analysis of Mel-spectrograms, attempts to encode these tonal differences and harmonic variations, which are the “key” to automatically bridging the semantic gap.

A. The Dataset

For the purposes of this research, the Emotify dataset [1] was used, which is available via the Kaggle platform. This specific dataset consists of 400 audio clips, each lasting 30 seconds.

1) *Collection Process and Crowdsourcing*: The data was collected through an innovative crowdsourcing game called Emotify, where volunteer listeners were asked to describe the emotions each track evoked in them [2]. This process differs from other methodologies as it focuses on the induced emotion (the emotion felt by the listener) rather than the perceived emotion (what the music tries to express), making the labels more human-centric.

2) *Musical Genres*: The sample is balanced and covers four primary music genres (Classical, Rock, Pop, Electronic), with 100 songs per genre. This diversity allows the model to train on different acoustic textures, from the calmness of orchestral pieces to the high energy of electronic music.

3) *Emotional Labels*: The primary dataset analyzed (data.csv) is characterized by a multi-entry structure per audio sample, reflecting the subjectivity of musical perception. Specifically, each of the 400 songs is not evaluated once but is repeatedly judged by different listeners, resulting in multiple records for the same track ID.

Emotional labels are assigned in a binary format (0 or 1), indicating whether the respective listener experienced one of the nine emotions of the GEMS-9 model [6]: amazement, solemnity, tenderness, nostalgia, calmness, power, joyful_activation, tension, sadness.

Each song can belong to more than one category simultaneously (multi-label classification), reflecting the complexity of the musical experience. This abundance of data per track allows for the subsequent statistical aggregation of the responses

to extract a representative “emotional map” for each song, forming the basis for training the CRNN.

II. METHODOLOGY

A. Data Preparation

The preprocessing procedure was divided into two main axes: the processing of emotional labels and the transformation of acoustic signals into a visual representation.

1) *Label Aggregation and Cleaning*: Due to the nature of the Emotify dataset, each song had multiple evaluations from different users. To avoid data leakage, the following steps were taken:

- **Aggregation**: Data was grouped by track ID. For each of the nine (9) emotions of the GEMS-9 model, the average of the binary responses (0 or 1) of all users who evaluated the specific track was calculated.
- **Binary Thresholding**: In order to extract the final labels for multi-label classification, a threshold mechanism was applied. After successive trials with values of 0.30 and 0.40, a stricter threshold of 0.50 was selected. For example, if the average response for an emotion (e.g., nostalgia) was ≥ 0.50 , the song was positively tagged for that emotion.

As observed in Figure 1, the distribution of the nine emotional classes presents significant class imbalance. For the evaluation of the baseline model, it was consciously decided not to apply any balancing techniques or class weights, to record the network’s objective generalization capability on raw data.

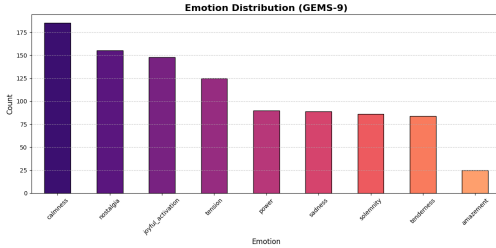


Figure 1. Distribution of the nine (9) initial emotional classes of the GEMS-9 model in the Emotify dataset. The intense imbalance (e.g., between Calmness and Tension) justifies the subsequent need for label reallocation and the introduction of Class Weights.

2) *Spectrogram Extraction*: The conversion of the audio signal into a Mel-spectrogram constitutes the stage where information is transformed from the time domain to the frequency domain, accounting for human psychoacoustics.

- **Mel-spectrograms**: Using the Librosa library [7], each sample was converted into a Mel-spectrogram with 128 frequency bands. The Mel scale was chosen as it corresponds to the non-linear perception of the human ear, which is more sensitive to changes in low frequencies than in high ones [8].
- **Log-Scaling (dB)**: Amplitude values were converted to a logarithmic scale (decibels). This is critical because it allows the model to “discern” patterns at low intensities

(e.g., the fading of a note in Calmness) that would appear as noise or empty space in a linear scale [9]. Logarithmic representation helps normalize the dynamic range, facilitating neural network training [10].

Figure 2 depicts the final input format for an electronic music sample. This visualization reveals the structure the CNN is tasked to recognize:

- 1) **Y-axis Frequency (Hz)**: Represents the 128 Mel bands. Low frequencies (bass, rhythm) are at the bottom, while highs (treble, synthesizers) are at the top.
- 2) **X-Axis Time**: Represents the temporal evolution of the clip (1292 time steps).
- 3) **Color Intensity**: The color gradient (from dark purple to bright yellow) indicates the magnitude of energy at each frequency. Strong vertical lines denote rhythmic “hits” (transients), characteristic of high-Arousal genres like Electronic and Rock.

3) *Input Normalization*: The final stage involved applying Min-Max Scaling. Spectrogram pixel values were normalized to the [0, 1] range. This action was necessary to stabilize the gradients during training, preventing the model collapse phenomenon observed in preliminary tests without normalization.

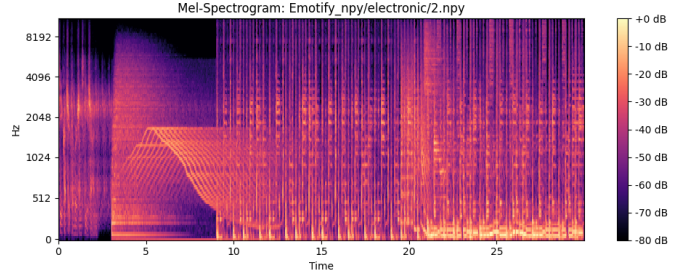


Figure 2. Log-Mel Spectrogram visualization of an electronic music sample. The 128 frequency bands and intense rhythmic activity indicating high energy levels (Energy/Joy) are clearly visible.

B. Initial Approach and Challenges (Baseline Model)

During the development of the Baseline CNN, the results highlighted the classic challenges of MER systems. Indeed, the Learning Curves revealed a sharp divergence between the training and test sets:

- **Accuracy**: While training accuracy exceeded 40%, test set accuracy stagnated near 15%.
- **Loss**: The characteristic “signature” of overfitting was observed, where after the 10th epoch, the test loss began to increase exponentially (over 1.0), while train loss continued to decrease.

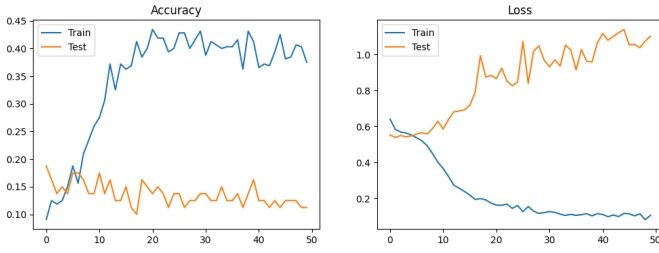


Figure 3. Learning curves of the initial CNN model. Severe overfitting is observed, as Test Accuracy remains low while Test Loss increases exponentially after the 10th epoch, indicating the model’s inability to generalize to new data.

This difficulty confirms the theory of Yang & Chen [5] regarding the **Semantic Gap**. The model became “trapped” in low-level technical audio features (noise, specific sample frequencies) instead of extracting high-level emotional information.

C. Final Architecture: Convolutional Recurrent Neural Network (CRNN)

Following the analysis of the initial model’s limitations, the hybrid CRNN architecture was adopted. This approach combines two different types of neural networks to fully comprehend the musical signal.

1) *Model Structure*: The model consists of three main segments:

- **Convolutional Segment (CNN Layers)**: Four Conv2D layers with an increasing number of filters (32, 64, 128, 256) were used. These layers act as “visual sensors” scanning the Mel-spectrogram to identify local features, such as percussion intensity or melody slope.
- **Recurrent Segment (RNN/GRU Layers)**: Following convolution, data is fed into Gated Recurrent Unit (GRU) layers. Unlike simple CNNs, GRUs allow the model to retain a “memory” of the temporal sequence. This is crucial for music, as an emotion (e.g., Tension) builds gradually over time.
- **Output Layer (Dense Layer)**: The final layer uses a Sigmoid activation function instead of Softmax, enabling multi-label classification. Thus, the model can recognize that a track evokes more than one emotion.
- **Dynamic Optimization and Control (Callbacks)**: To ensure smooth convergence and prevent overfitting, training was enhanced with dynamic mechanisms. Initially, Early Stopping was applied, automatically restoring the best weights if validation loss showed no improvement for 10 consecutive epochs. Furthermore, to avoid entrapment in local minima, adaptive learning rate adjustment (Reduce LR on Plateau) was integrated. This mechanism reduced the initial learning rate (0.0003) to 20% of its value when the loss remained stagnant for 4 epochs, allowing the network to make finer adjustments as it approached optimal convergence.

Despite the theoretical superiority of the hybrid CRNN architecture in processing music signals, the first application

of the model to the Emotify dataset did not yield the expected results. Despite the network’s ability to locate spatial and temporal features, training still encountered the problem of insufficient generalization. As depicted in the following learning curves, the model exhibited a strong tendency to overfit, confirming that architectural complexity alone is not enough to overcome the limitations of a small and imbalanced dataset.

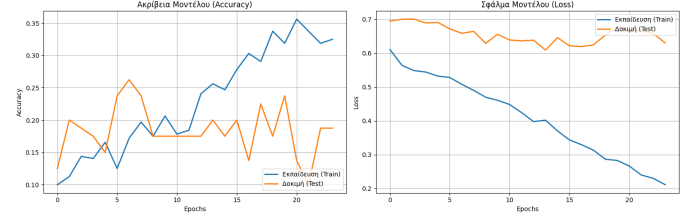


Figure 4. Learning curves of the initial CRNN implementation. Severe overfitting is again observed; although Test Accuracy increases slightly, it remains low, while Test Loss stays consistently high (between 60% and 70%). The model’s inability to generalize remains.

2) *Overfitting Mitigation Techniques*: To ensure the model’s generalization across the 400 songs in the dataset, the following mechanisms were incorporated:

- **Batch Normalization**: Applied after each convolution to stabilize training and accelerate convergence.
- **Dropout (0.3 - 0.5)**: Random neuron dropout layers were introduced to force the network to learn more robust features and avoid memorizing dataset noise.
- **Global Average Pooling**: Replaced traditional Flatten layers, drastically reducing the number of parameters and the risk of overfitting.

3) *Label Reallocation*: The original nine GEMS-9 labels exhibited significant overlaps in a multi-label environment. To enhance the model’s discriminative ability and align with the needs of the mental health application, the labels were reallocated into four main groups. Following the methodology of Yang & Chen [5], emotions were mapped onto the two-dimensional Valence-Arousal space.

- 1) **Energy/Joy**: Consolidated the Power, Joyful Activation, and Amazement labels, focusing on high arousal with positive valence.
- 2) **Calm/Relax**: Merged the Calmness, Tenderness, and Solemnity categories, representing low arousal states with positive valence.
- 3) **Sadness/Nostalgia**: Formed by combining Sadness and Nostalgia, covering the spectrum of melancholic emotions (low arousal with negative valence).
- 4) **Tension/Fear**: Based on the Tension label, representing high arousal with negative valence.

This reallocation reduced the complexity of the problem, allowing the CRNN model to achieve higher Accuracy and more robust generalization, as the new classes are more clearly separated in the acoustic feature space.

4) *Handling Data Imbalance (Class Weighting)*: Upon analyzing the sample distribution in the dataset, a significant

class imbalance was observed. Specifically, while Energy/Joy, Calm/Relax, and Sadness/Nostalgia each had approximately 150 samples, Tension/Fear had only 78. To prevent the model from biasing toward the more frequent classes, the Class Weights technique was applied during training, utilizing the Inverse Frequency method:

$$W_i = \frac{N_{total}}{M \cdot N_i} \quad (1)$$

Where N_{total} is the total number of samples, M is the number of classes, and N_i is the number of samples in class i . Based on this formula, the following weights were assigned:

- **Energy/Joy:** 0.91
- **Calm/Relax:** 0.84
- **Sadness/Nostalgia:** 0.88
- **Tension/Fear:** 1.70

The increased weight (1.70) in the Tension/Fear category forces the model to assign greater penalty to errors regarding this class, ensuring the CRNN develops the necessary sensitivity despite the lack of abundant samples.

5) *Dataset Augmentation:* Due to the limited size of the Emotify dataset (400 songs), training a Deep Learning network was expected to lead to overfitting. To counter this, data augmentation techniques were applied to the acoustic features to artificially increase sample diversity:

- **Add Noise:** Light white noise (Gaussian noise) was introduced to the spectrograms so the model could learn to recognize dominant musical patterns even in “unclean” signals.
- **Time Shifting:** A slight shift of the spectrogram along the time axis was performed, ensuring the model does not rely on the absolute temporal position of a musical event (e.g., a drum hit) to extract emotion.

The combined application of these techniques allowed for the artificial multiplication of available training data, effectively tripling the dataset size.

III. RESULTS & DISCUSSION

The evaluation of the final model was conducted on the test set, which includes samples the network had not encountered during training. The adoption of the CRNN architecture combined with normalization techniques led to a significant improvement in accuracy, reaching 80.3%.

A. Learning Curves and Convergence

Analysis of the learning curves proves that interventions in architecture and data preprocessing stabilized the training process. As shown in Figure 5, the divergence between training and validation loss is minimal. The model possessed the necessary diversity to learn generalizable musical features.

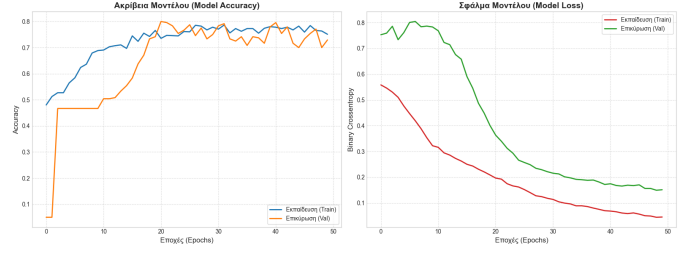


Figure 5. Accuracy (left) and Loss (right) curves of the final CRNN model. Smooth convergence and elimination of overfitting are observed.

B. ROC Analysis and Discriminative Ability

The use of Class Weights (with the characteristic weight of 1.70 for the Tension class) allowed the model to maintain high performance across all categories, despite the initial data imbalance.

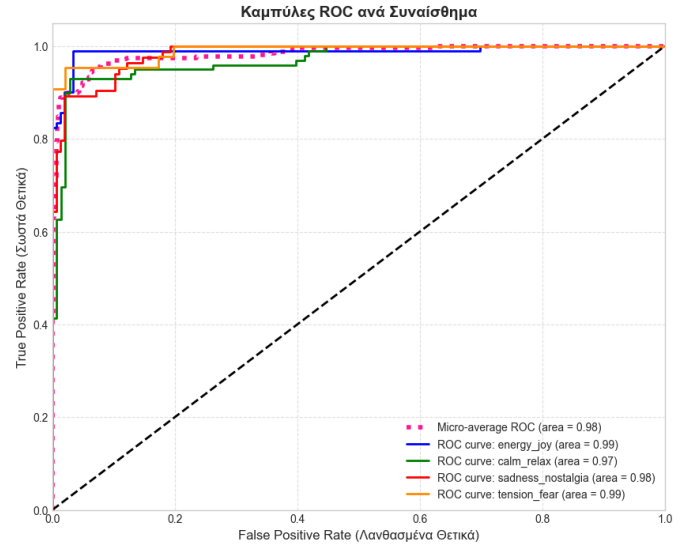


Figure 6. ROC curves for the four emotional classes.

In Figure 6, AUC (Area Under Curve) values indicate excellent discriminative ability. Particularly in the Energy/Joy and Calm/Relax categories, the model achieves near-optimal classification, which is attributed to the GRU layers’ ability to encode the temporal evolution of rhythm and harmony.

C. Best Model Selection (Epoch 49)

The final evaluation was based on the weights extracted during the 49th epoch. The selection arose from observing the loss curves, where Validation Loss stabilized at its minimum value before exhibiting overfitting trends.

Results on the test set for Epoch 49 are impressive, as shown in Table I:

Emotional Class	Precision	Recall	F1-score	Support
Energy/Joy	0.96	0.87	0.91	91
Calm/Relax	0.95	0.93	0.94	99
Sadness/Nostalgia	0.93	0.89	0.91	84
Tension/Fear	0.93	0.91	0.92	43
Macro Avg	0.94	0.90	0.92	317

Table I
PERFORMANCE METRICS PER CLASS (EPOCH 49)

This performance (weighted avg F1-score 0.92) is remarkably high for the field of MER. Noteworthy is the excellent balance in the Tension/Fear class (F1 0.92), which, despite having the fewest samples, truly benefited from the class weight tuning.

The high Precision value (up to 0.96) ensures that the supportive mental health interventions proposed by this work will rely on highly reliable predictions, minimizing false positives.

Figure 7 presents the Confusion Matrices for each of the four emotional classes. Due to the nature of the problem (multi-label classification), evaluation is performed adopting a One-vs-Rest strategy. A similar ratio of correct predictions is observed for each emotion, with the highest accuracy found in the Tension/Fear class, attributed to class weight tuning.

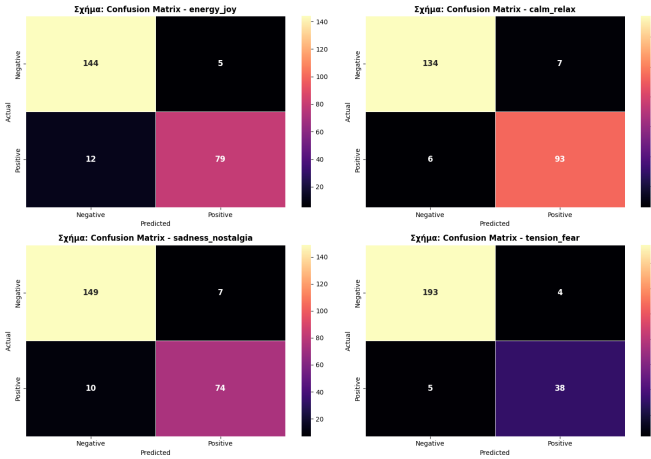


Figure 7. Confusion Matrix of the predictions.

D. Qualitative Error Analysis

While Figure 7 provides a quantitative depiction of errors at the class level, a more thorough, qualitative analysis of the model's raw data was deemed necessary to understand its behavior. Analyzing a sample of 36 incorrect predictions, beyond 4 isolated failures, three dominant patterns were identified:

- 1) **Hesitation Errors:** 14 borderline cases that did not cross the decision threshold due to the CRNN model's conservative behavior.
- 2) **Confusion between Calm/Relax & Sadness/Nostalgia:** 8 cases where arousal levels are conflated because these two emotional states share similar low arousal characteristics.
- 3) **Failure to Detect Energy/Joy:** 10 cases where the GRU layers failed to track intensity escalation and potentially

over-focused on the initial acoustic information of the track.

E. Effect of Musical Genre

The final phase of the evaluation involved checking predictions relative to the musical genre. According to Figure 8:

- Classical music largely identifies with calmness (Calm/Relax).
- Electronic music shows high percentages in the Energy/Joy and Tension/Fear categories, likely due to fast tempos, synthesizers, and bass (high arousal).
- Rock is distributed across all categories as it encompasses many sub-genres. The small percentage in the Tension/Fear category possibly indicates a lack of Metal tracks in the sample.
- Pop similarly exhibits calmness, with higher amounts of Sadness/Nostalgia and Tension/Fear, presenting a more melancholic overall profile.

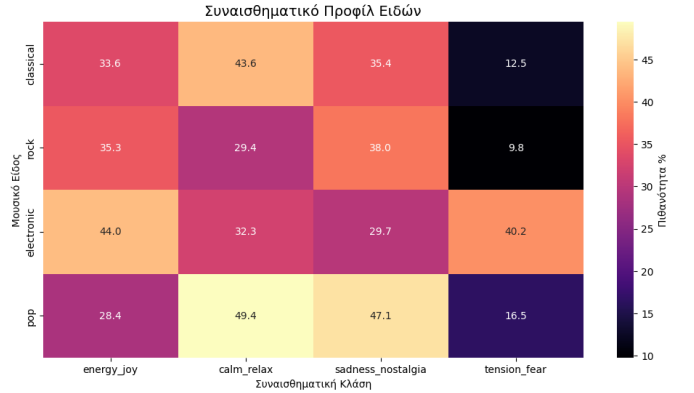


Figure 8. Distribution of emotional predictions per musical genre.

This finding is in full agreement with the literature of Aljanaki et al. [3], proving that the model successfully bridged the Semantic Gap by linking spectrograms to human perception. Overall, the combined analysis of metrics and genre distribution confirms that the CRNN architecture can extract valid emotional information, making it a reliable tool for automated mental health support.

IV. APPLICATION IN MENTAL HEALTH

The need for automatic detection of emotional trends is grounded in studies linking extensive listening to sad music with negative psychological effects under specific conditions. According to Garrido and Schubert [11], while sad music can offer relief to healthy listeners, for individuals prone to rumination—continuous engagement with negative thoughts—prolonged exposure to such music acts as a maintenance mechanism for depressive moods. This view is corroborated by Skånland's research [12], which points out that the use of personal music players might lead to “emotional trapping,” where the user becomes isolated in a bubble that amplifies their sadness. The application intervenes exactly at this point, utilizing the predictive capabilities of the CRNN

model to break this vicious cycle and suggest content that promotes mood regulation [13].

A. Playlist Analysis Mechanism

Instead of isolated song analysis, the application processes entire folders or user playlists. The procedure includes:

- **Batch Processing:** The system scans .mp3 files, extracts Log-Mel Spectrograms, and generates probabilistic predictions for each track.
- **Cumulative Evaluation:** The final prediction for each song is derived not from a single sample, but from the average of the predictions of all its segments. This reduces statistical error and highlights the true emotional identity of the track.
- **Secondary Trend Analysis:** For a more complete mapping of mood, the system does not restrict itself to the dominant class but also detects secondary trends, provided they exceed a Confidence Threshold of 30%. This allows the identification of complex states, such as “Bittersweet” or “Numbness”.
- **Emotional Profile:** The average of the emotions across the entire playlist is calculated, creating a “map” of the user’s current psychological state.

B. Alert and Intervention System

Based on the analysis of the above data, the system triggers personalized Mental Health Prompts. The intervention logic was coded to react to specific combinations:

- 1) **Emotional Rumination Detection:** In cases where playlist analysis shows a high convergence of Sadness and Calm categories in repeated tracks, the application issues a warning for potential emotional trapping, encouraging the user to change their listening choices.
- 2) **Acute Tension Management:** When the Tension/Fear category is combined with high percentages of Sadness, the system recognizes a state of agitation and suggests an immediate transition to Calm/Relax content.

As shown in Figure 9, the application visualizes these results, offering immediate feedback to the user.

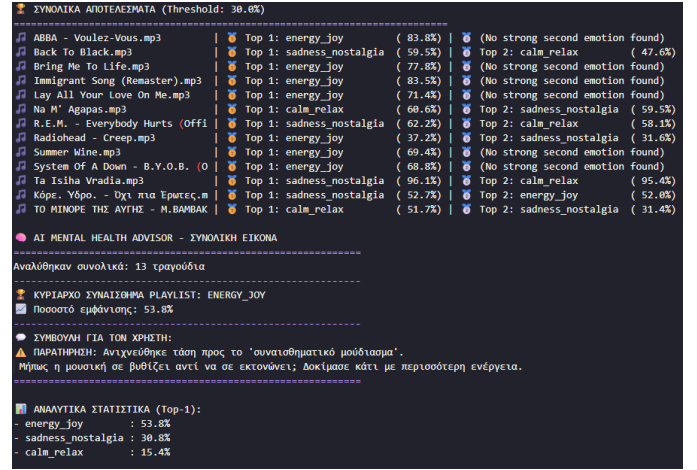


Figure 9. Command-line interface screen of the AI Mental Health Advisor prototype system. The image captures the execution of the algorithm during the batch analysis of a playlist. It highlights individual predictions (Top-1 and Top-2) per track, the calculation of the average emotional profile, and the automatic generation of a personalized supportive message. This specific visualization is provided to demonstrate the full functionality and practical application of the theoretical model in real-world operating conditions.

V. CONCLUSIONS

This work demonstrated that deep learning can effectively bridge the gap between digital signals and human psychology. The CRNN architecture utilized is considered state-of-the-art for acoustic data analysis, as it combines the ability of CNNs in extracting spatial features from spectrograms with the ability of GRUs to perceive the temporal evolution and rhythm of music.

The model’s high accuracy (0.92 F1-score) enables the creation of reliable tools that can function as companion applications in digital psychotherapy and self-help platforms.

Finally, it should be noted that even if an algorithm achieves a theoretically perfect accuracy rate (100%), its nature remains purely advisory. The complexity of the human psyche cannot be confined to a computational model, and therefore, the application does not constitute a mental health authority, but an assistive and preventive tool. Concurrently, this research highlighted that emotion recognition via audio requires further improvement, particularly in understanding emotional ambiguity and subtle nuances that only human perception can fully grasp. This admission does not diminish the system’s value but lays the groundwork for the future enhancements described below.

VI. FUTURE EXTENSIONS

Analyzing exclusively audio presents inherent limitations, as it does not account for the semantic content of the lyrics or the listeners’ subjective experience. A proposed improvement for the evolution of the application is the adoption of a multifactorial approach.

A. Lyrics Analysis

The automatic retrieval of lyrics and their analysis for emotional valence (Lexical Sentiment Analysis) is the first

step in filling the void left by melody. Many musical works are often characterized by high acoustic energy which the model might misinterpret as “joy,” while the lyrics convey deeply melancholic messages. Integrating text-analyzing algorithms would allow the system to recognize the thematic content and weight the final prediction accordingly, offering a more accurate emotional identification.

B. Integration of Social Data and Comment Analysis

A promising extension is the integration of social metadata and the analysis of user interaction with each work. Leveraging Natural Language Processing models to analyze user comments, as well as tag processing, can act as a robust confirmation system.

C. Connection with Music Streaming Platforms

The practical utilization and scalability of the proposed system can be maximized through its integration with popular music streaming services. By utilizing APIs, the application would be able to analyze the user’s listening habits in real-time. Thus, it would intervene dynamically, reconfiguring the current playlist with the goal of smoothly relieving the listener’s emotional burden.

ARTIFICIAL INTELLIGENCE USAGE DECLARATION

Generative artificial intelligence tools were used for language editing, LaTeX formatting, and support in the debugging process of the CRNN model. The conception of the research idea, the architectural design, the training of the model, as well as the analysis of the results, are exclusively the product of the author’s personal research work.

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